
Local-first Software

Prof. Dr. Annette Bieniusa

AG Software Technology



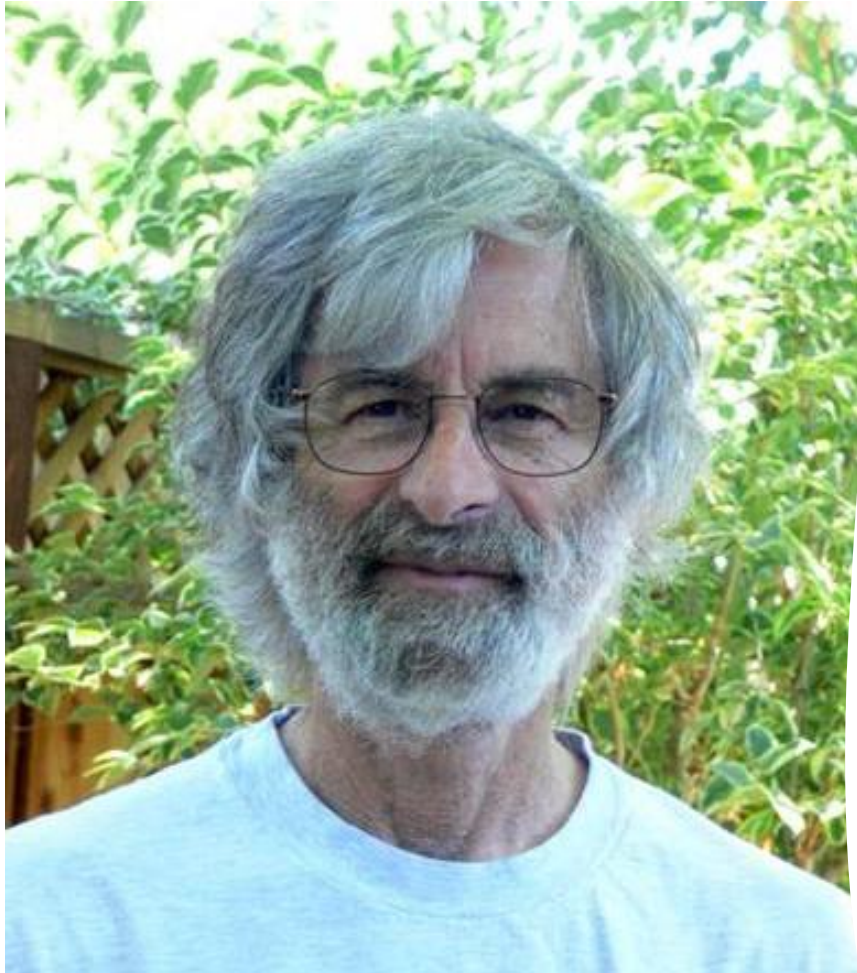
Are you developing an application that allows users to share data?

Does it work if the server goes down?



Does it work if you loose the internet connection?





„A distributed system is one in which the failure of a computer you didn't even know existed can render your own computer unusable.“

Leslie Lamport
(1987)

RPTU

The tale of Klara and Marie

Let's organize a workshop!



How did users typically collaborated 10 years ago*



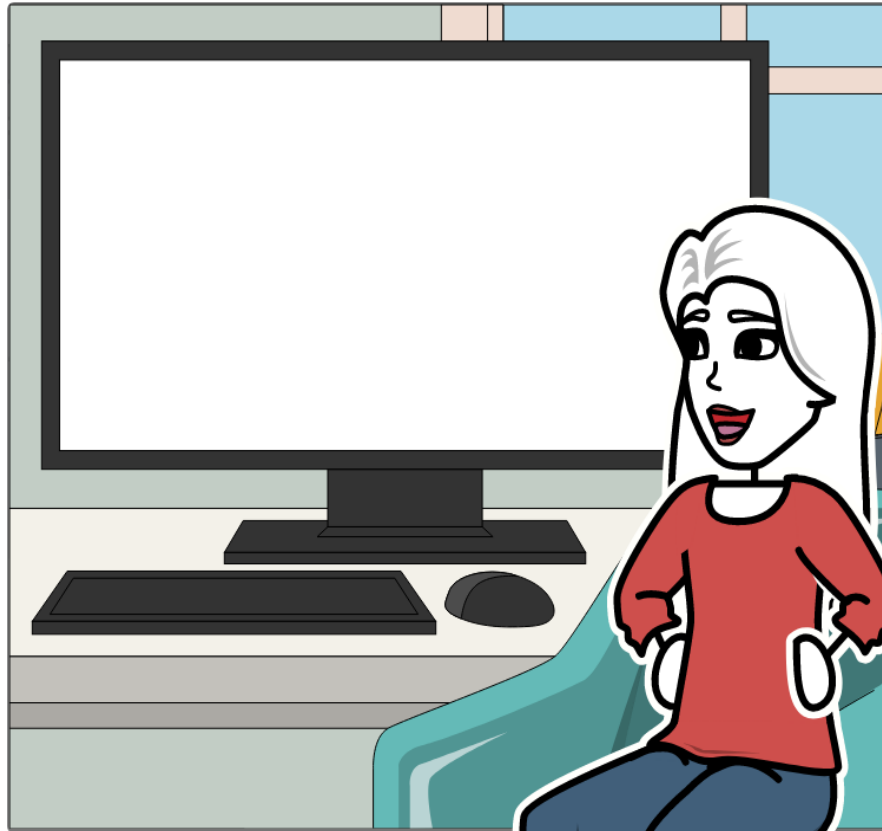
Dear Marie,

Enclosed is my first draft for the schedule!

Kind regards,
Klara

 BOB_Workshop.docx

How did users typically collaborated 10 years ago*



Dear Klara,

Great, I've added my ideas.

Best regards, Marie

 BOB_Workshop_modified_by_Marie.docx

How did users typically collaborated 10 years ago*



Dear Marie,

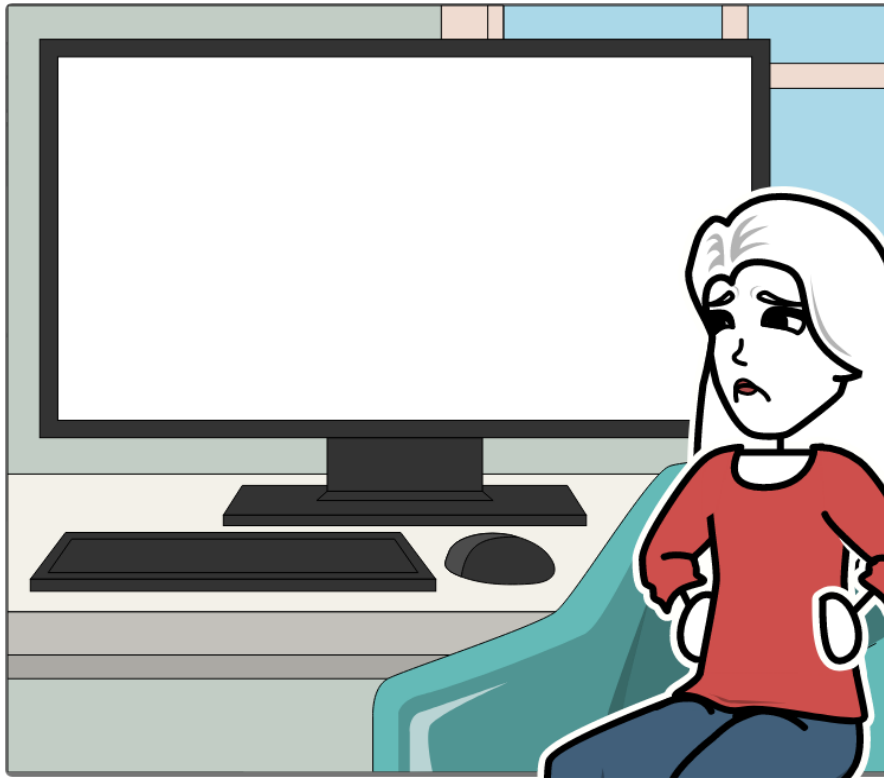
I have adapted task 4.

Kind regards, Klara



BOB_Workshop_Marie_Version_24.July.docx

How did users typically collaborated 10 years ago*



Oh crap, I changed the task too...



BOB_Workshop_modified_by_Marie(2).docx

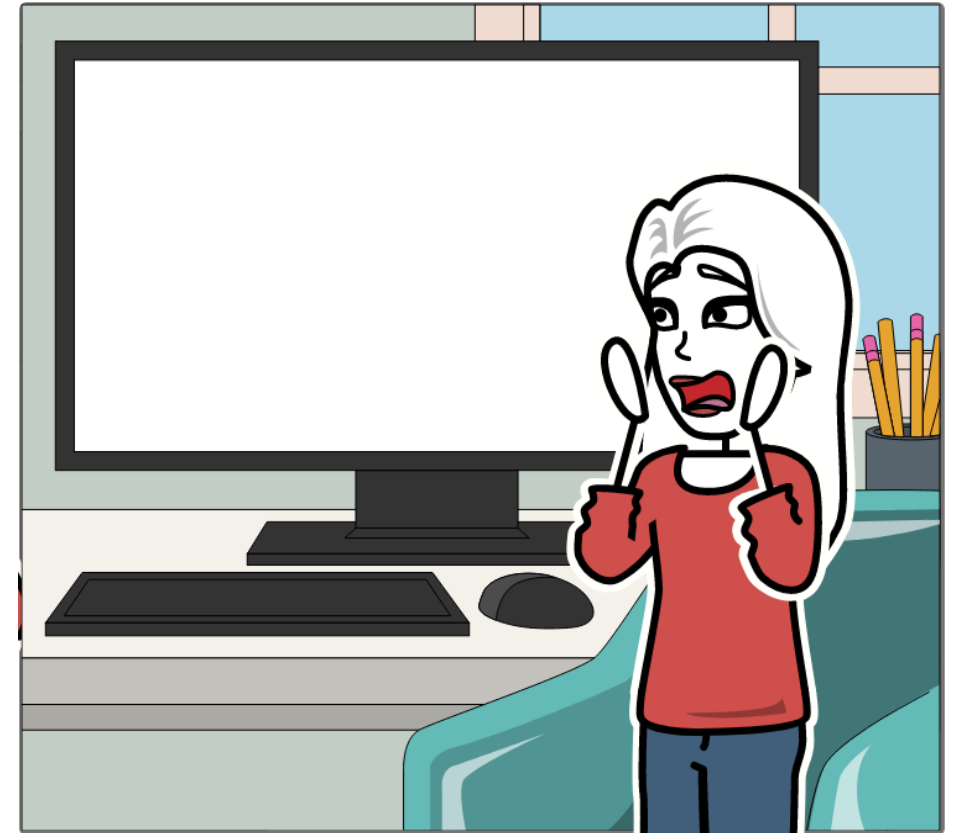
The result

 BOB_Workshop.docx

 BOB_Workshop_modified_by_Marie_again.docx

 BOB_Workshop_Marie_Version_24.July.docx

 BOB_Workshop_modified_by_Marie(2).docx



RPTU

Today: Use some cloud service to manage the document

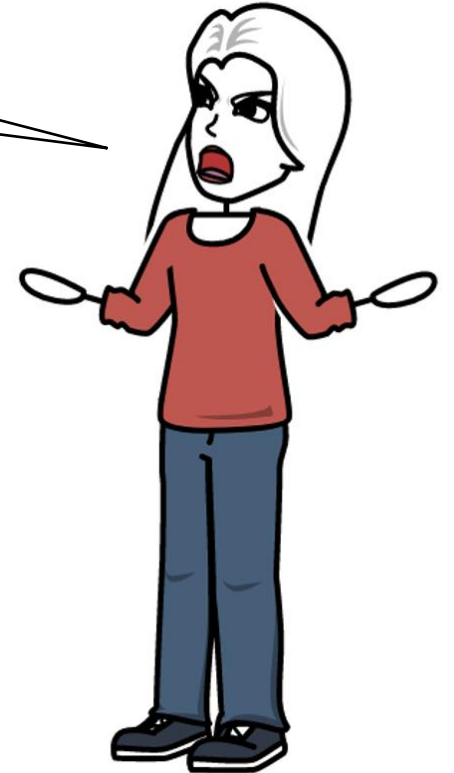


Now, everybody can edit at any time



Where does
this file come
from?!?

Name		Date Modified	
	Sample File		Today at 10:54
	Sample File (hanz-mbp's conflicted copy 2018-10-12)		Today at 10:54



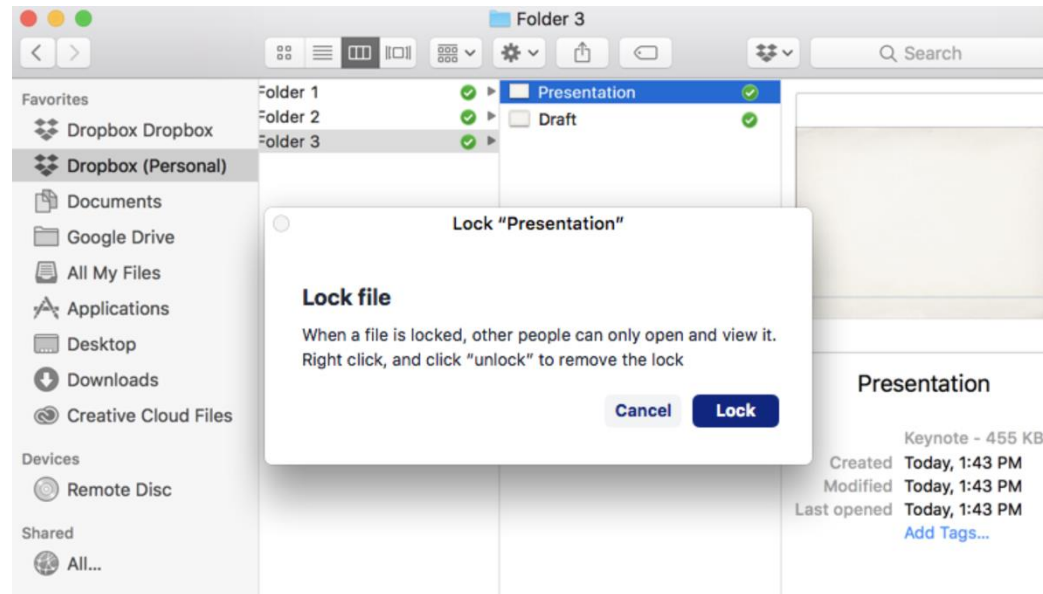
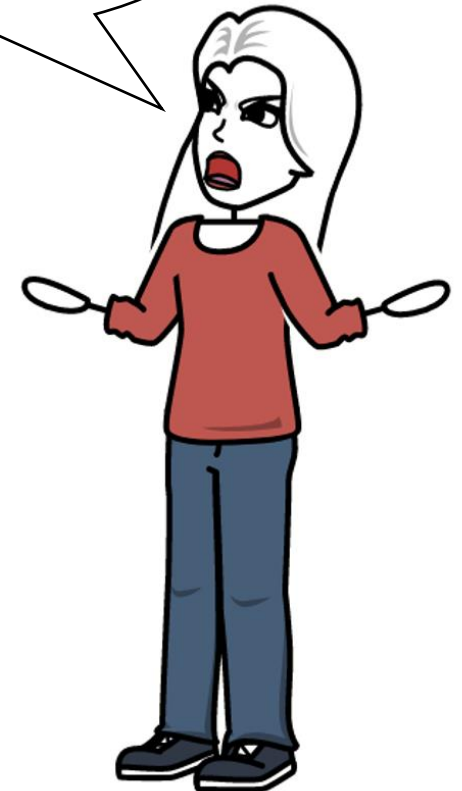
RPTU

Option: Enforce Mutual Exclusion



Hmm, I would like to revise the text.
How can I make sure that Marie
doesn't get in my way?

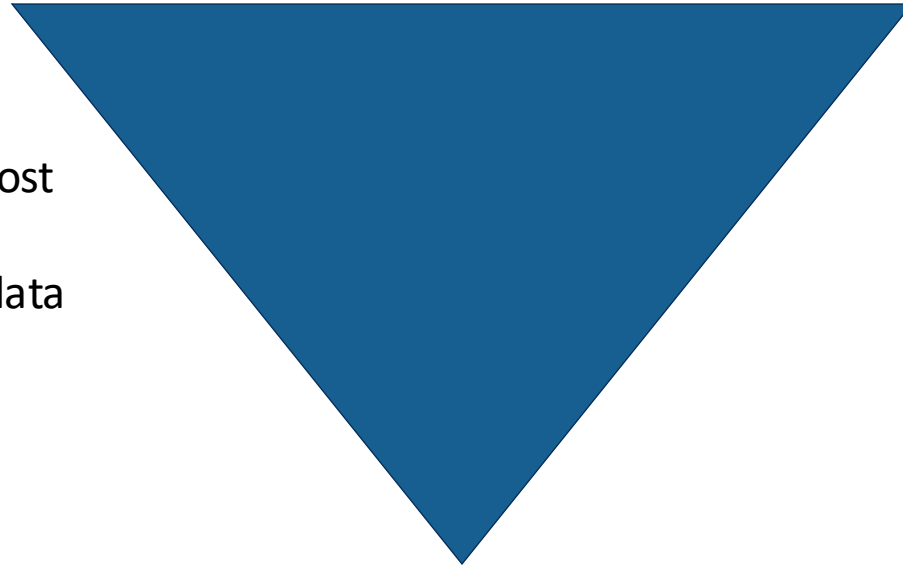
Why can't I open
the file?



CAP Theorem [Brewer '00, Gilbert/Lynch '02]

Consistency

Every read receives the most recent write or an error.
All nodes have the same data at the same time.



Availability

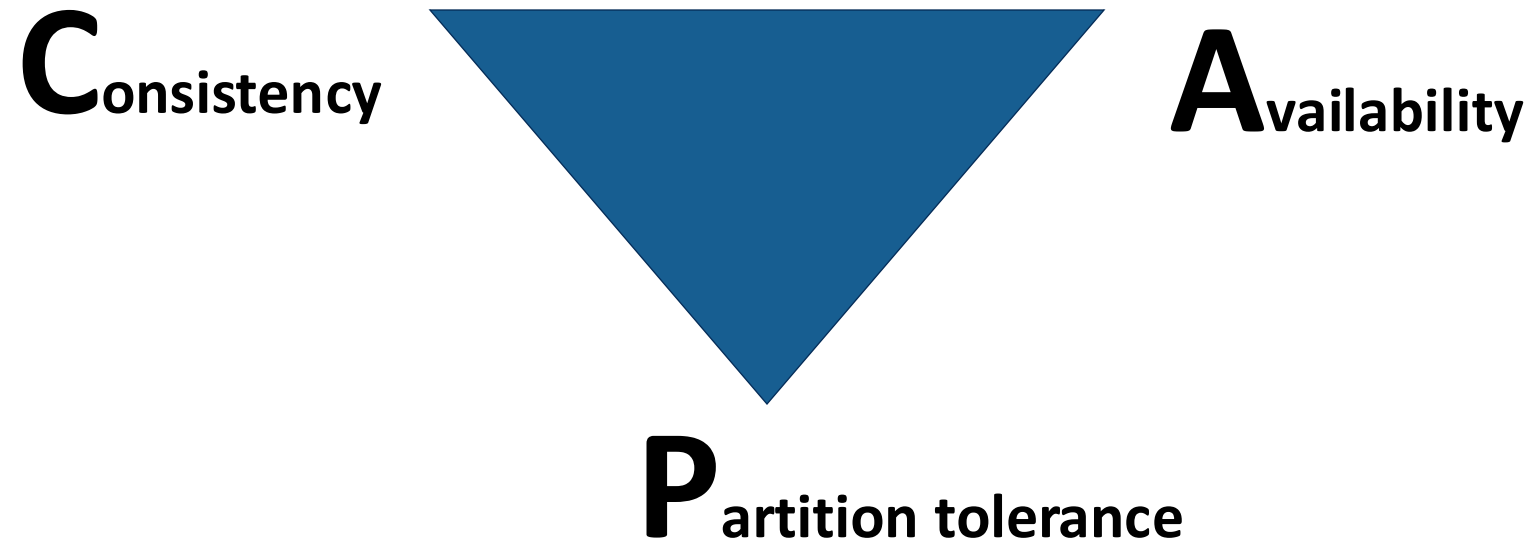
Every request (read/write) receives a response, even if some nodes are down.

Partition tolerance

The system continues to operate even when network failures (partitions) occur between nodes.

RPTU

CAP Theorem [Brewer '00, Gilbert/Lynch '02]



A distributed data store can only provide
two out of the three properties at the same time.

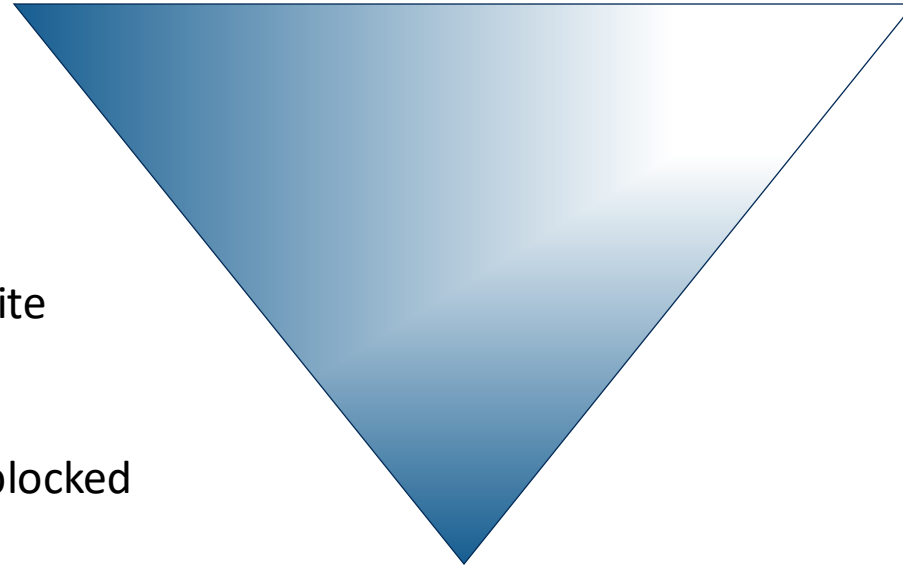
CAP Theorem [Brewer '00, Gilbert/Lynch '02]

Consistency

- Requires coordination of write
 - Mutual exclusion
 - Consensus
- Reads and writes might be blocked

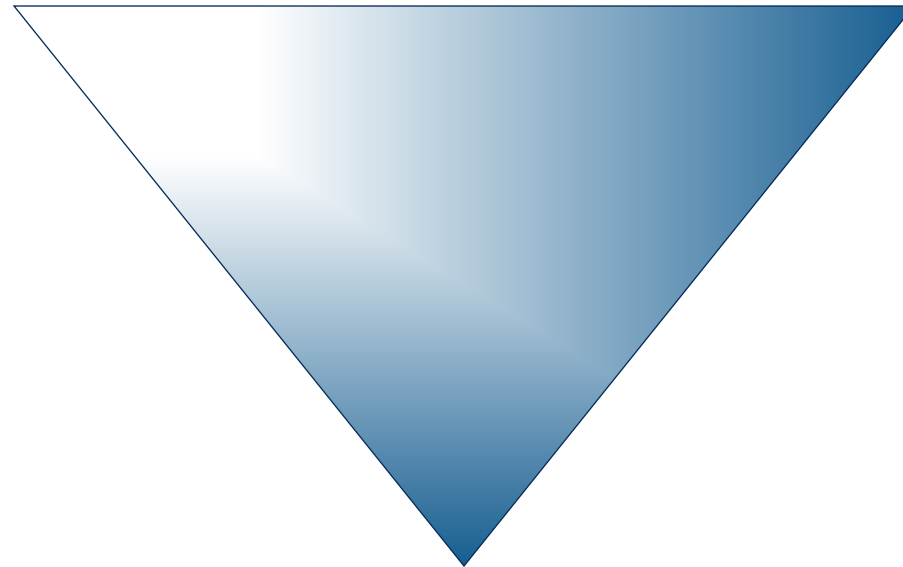
Availability

Partition tolerance



CAP Theorem [Brewer '00, Gilbert/Lynch '02]

Consistency



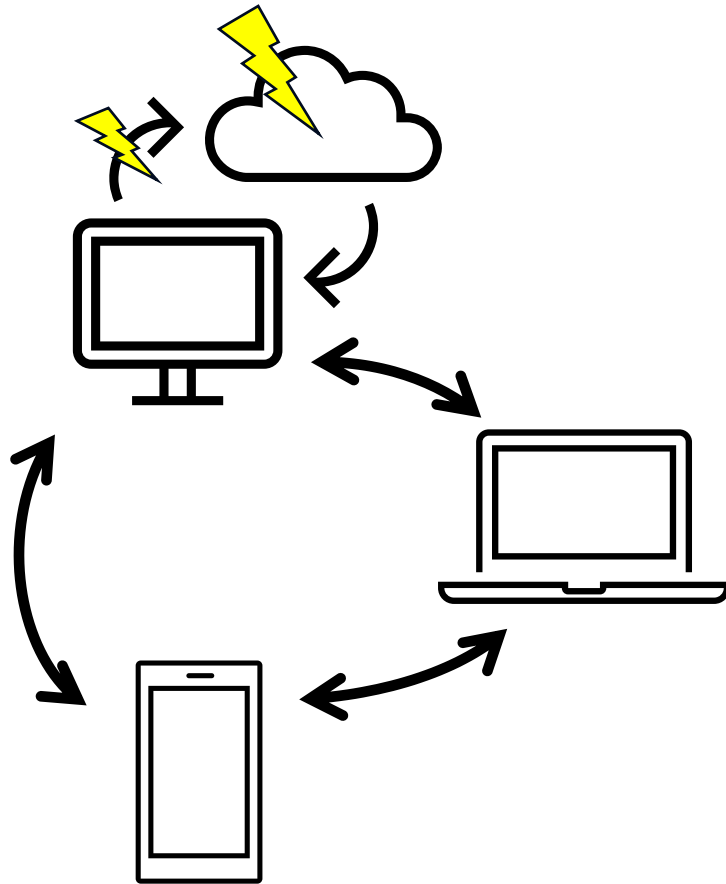
Availability

- Requires conflict resolution
- Reads might observe updates only eventually

Partition tolerance

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Local-first Software



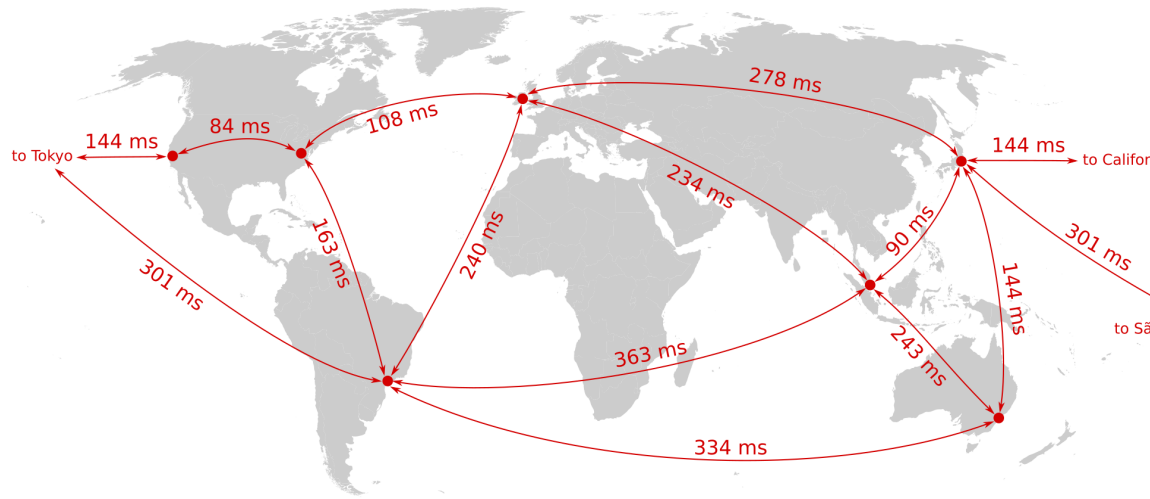
„In local-first software, the availability of another computer should never prevent you from working.“

Martin Kleppmann
(Local-first Conference, Berlin 2024)

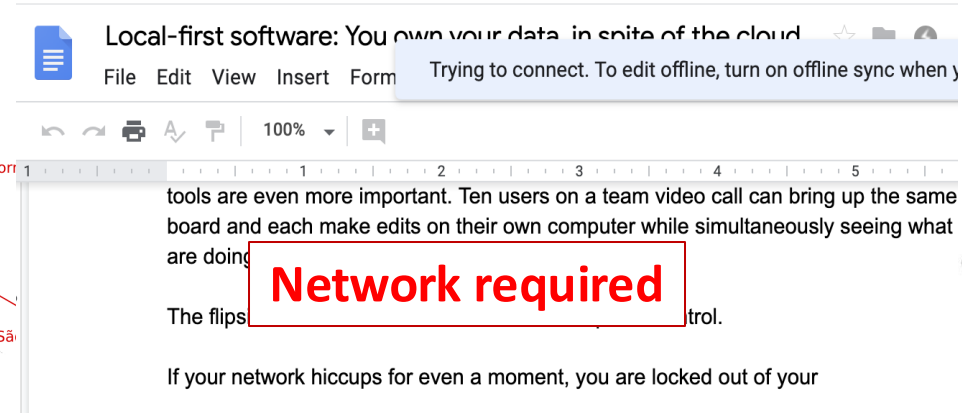


Principles of Local-First Software

Synchronisation via servers can be problematic



Latency



Dependence on service providers

- **Lifetime of service**
- **Business model**
- **Data ownership**



Local-first Software Manifesto

1. No spinners: Your work at your fingertips
2. Your work is not trapped on one device
3. The network is optional
4. Seamless collaboration with your colleagues
5. The Long Now
6. Security and privacy by default
7. You retain ultimate ownership and control

Martin Kleppmann, Adam Wiggins, Peter van Hardenberg, and Mark McGranaghan.

Local-first software: you own your data, in spite of the cloud.

2019 ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software (Onward!), October 2019

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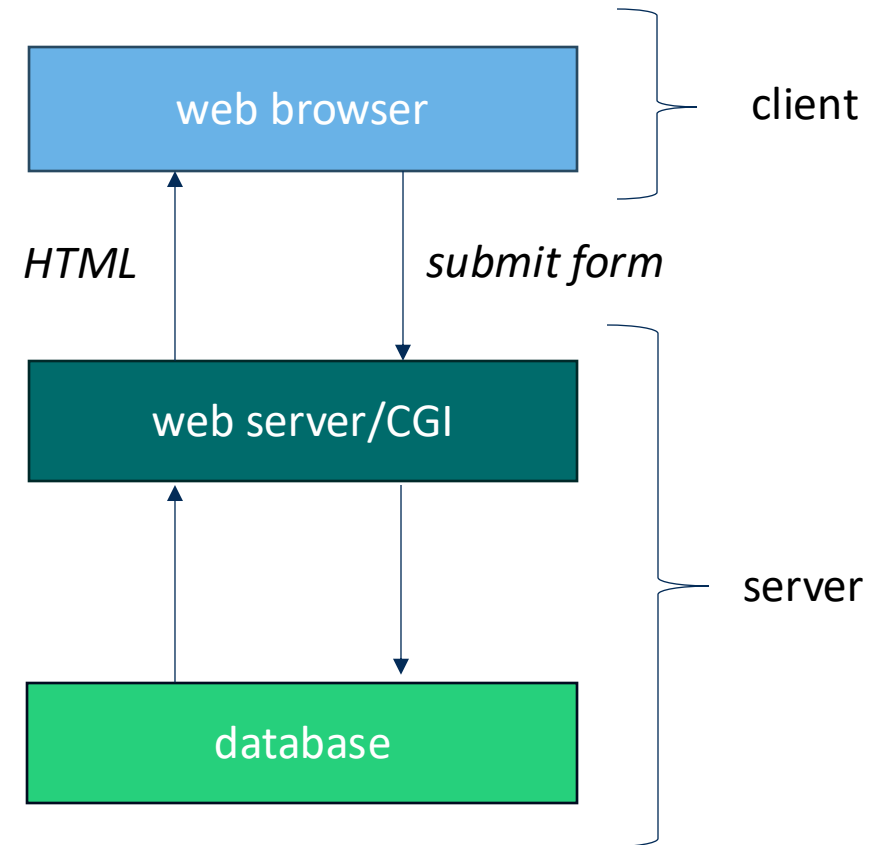
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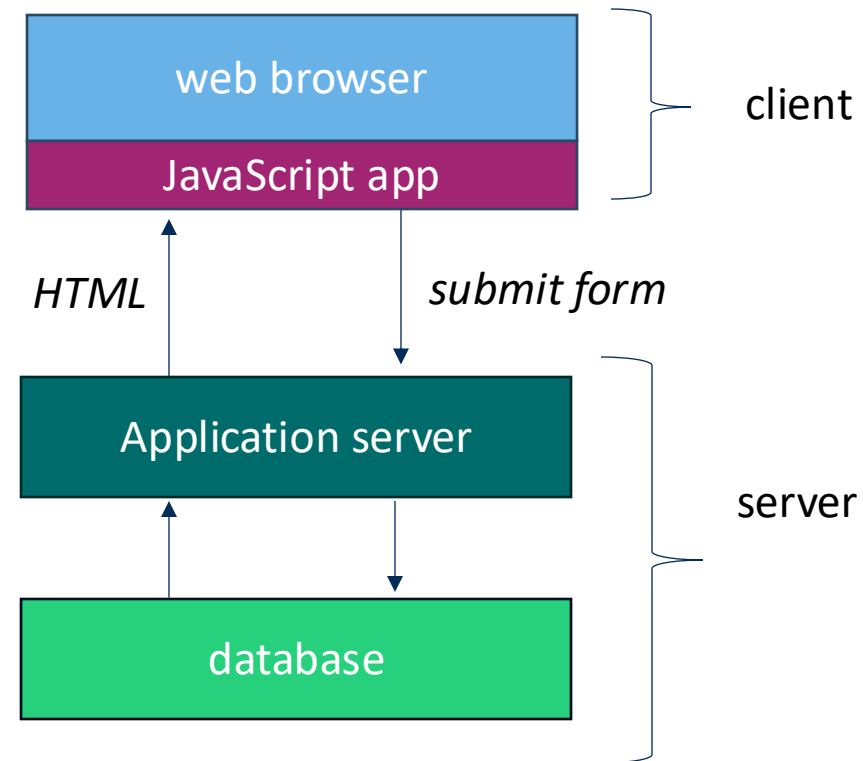
Web applications over the course of time

In the early days (ca. 1990 - 2000):



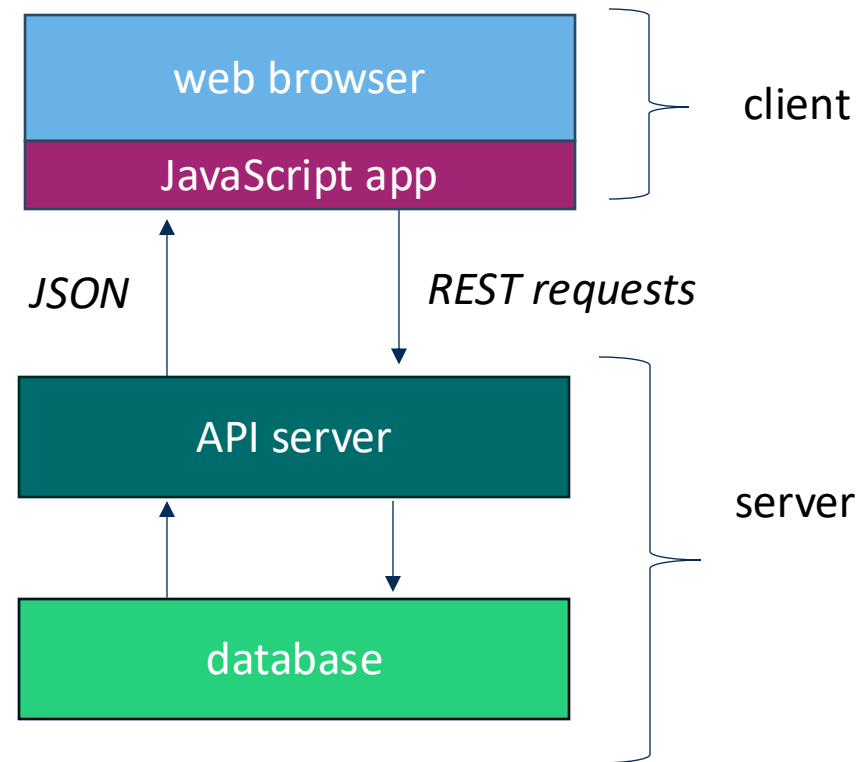
Web applications over the course of time

In the next decade (ca. 2000 - 2010):

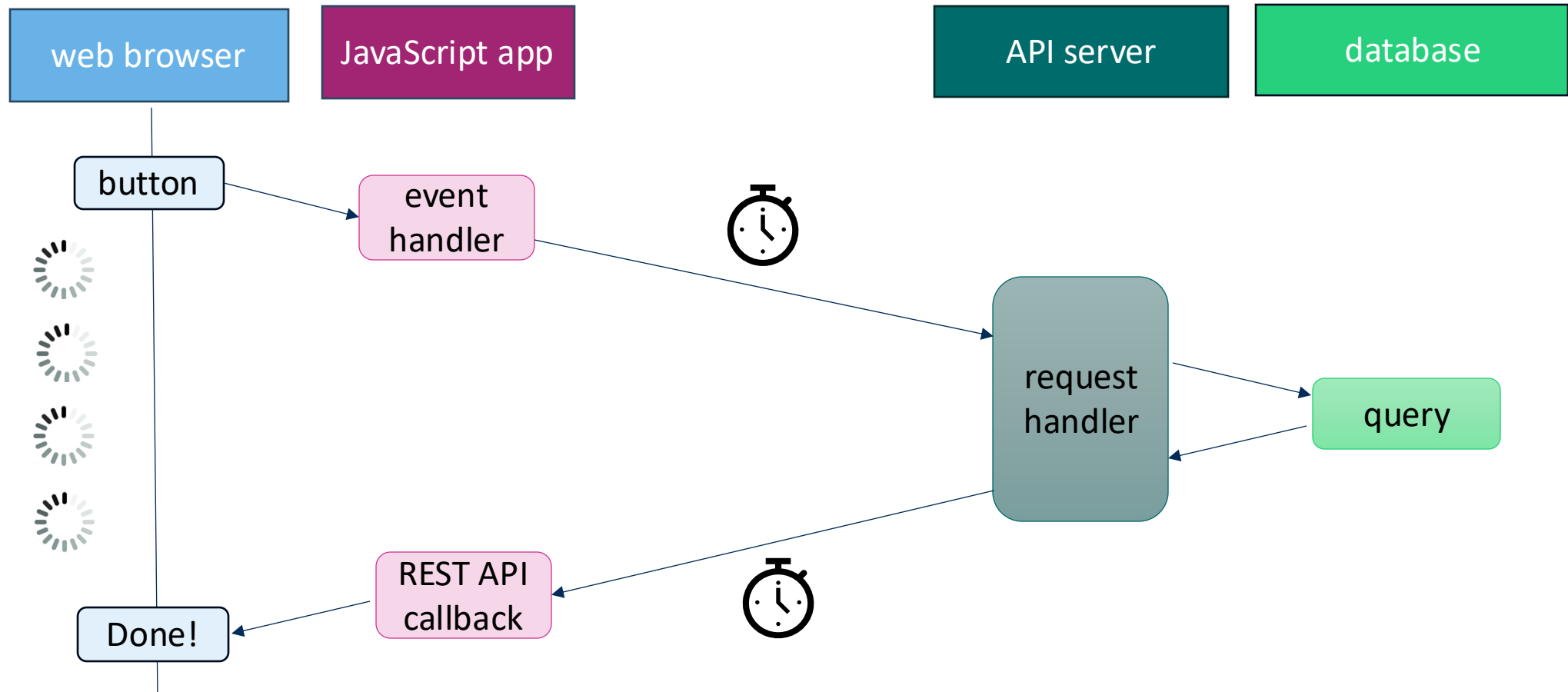


Web applications over the course of time

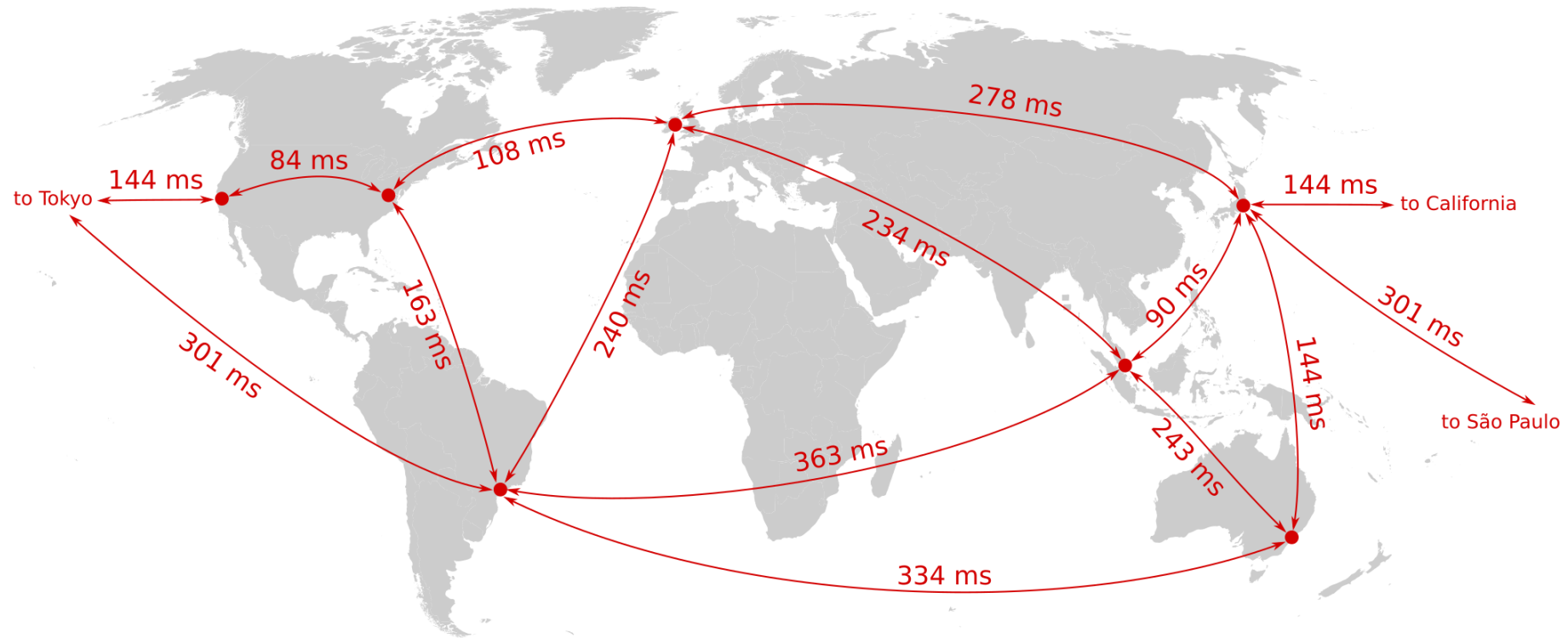
More recently (ca. 2010 - 2020):



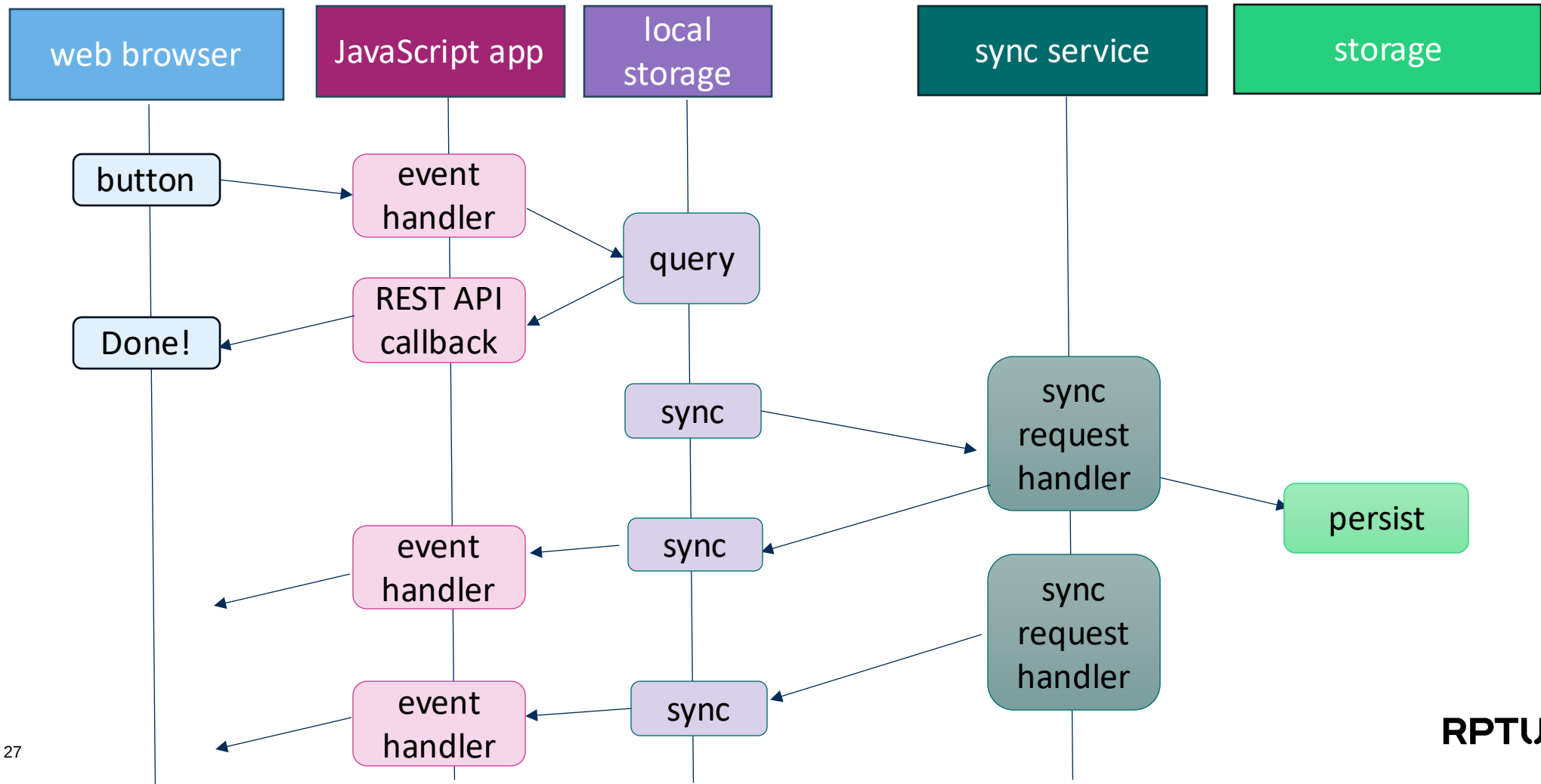
When we click a button...



The Internet is slow



What we actually want



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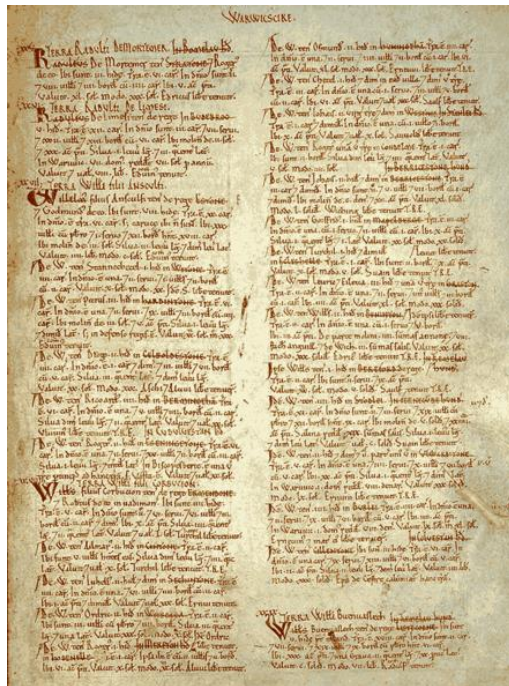
The Long Now



Early written tablet recording the allocation of beer, probably from southern Iraq, late prehistoric period, 3100-3000 BC

Source: WikiCommons

From paper to silicon: The Domesday Project

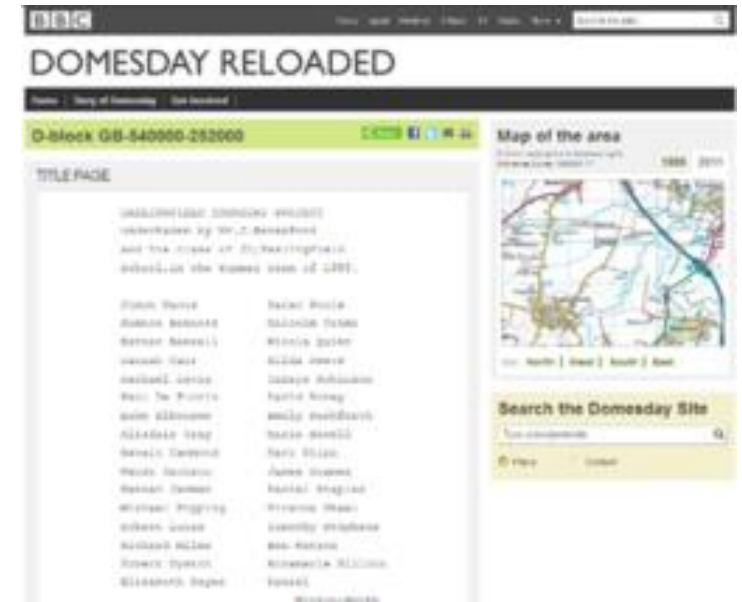


https://en.wikipedia.org/wiki/Domesday_Book

11th century: Land register of England documenting the legal ownership of land



1986: Multimedia Edition with information about modern Britain; accessible only on BBC Master Advanced Interactive Video System

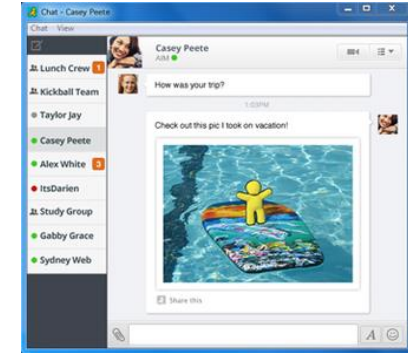


2011-2018: Online access for the digitised and updated content; transferred to The Nationale Archives

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Escaping the Digital Dark Ages

- Open Standards (XML, JSON, SQLite)
- Redundant and replicated storage
- But what about the application software?



and many, many more...



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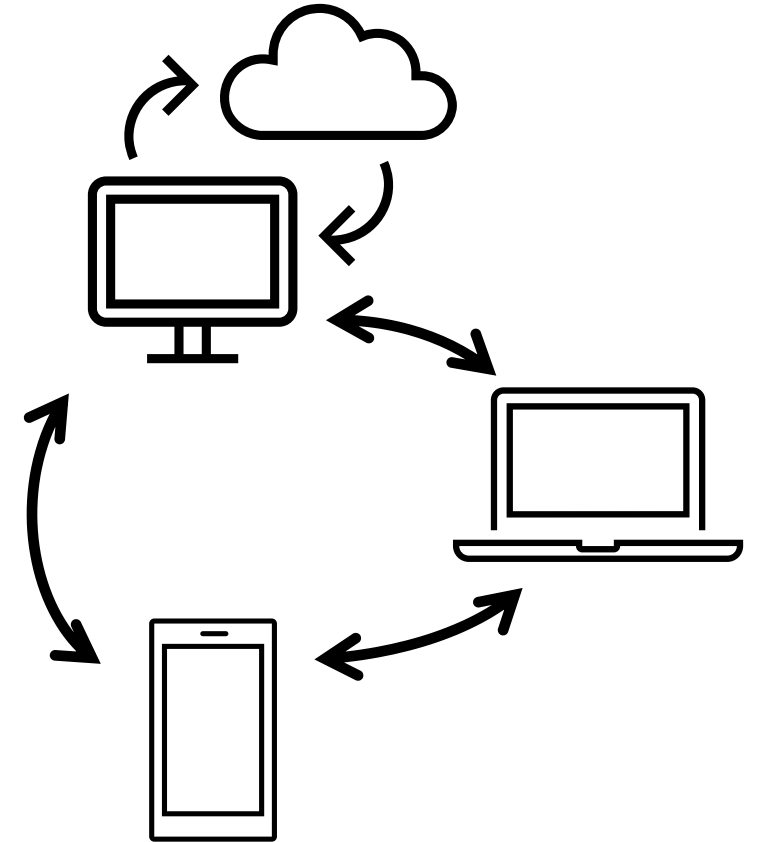
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Collaboration with other users

- Working concurrently with other users
- Many possible applications
 - Office applications: Text documents, presentations, ...
 - Project management tools: To-Do lists, calendars, ...
 - Games
- Reactive: No waiting for the server
- Safe and secure



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Techniques for Local-First Software

Concurrent updates to data can have interesting effects



@thenizami

Follow



How tf do you have a negative number of followers, what kind of voodoo is this fam
[@AnassLamouri](#)



+ Follow

Anass Lamouri BS.

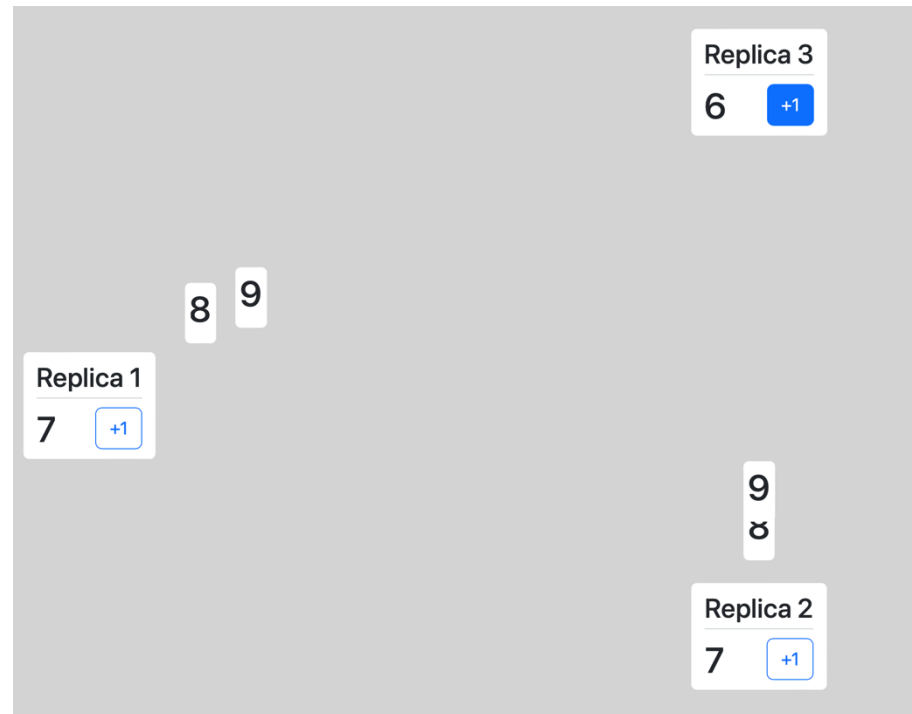
@AnassLamouri

|BANLIEUE SALE MUSIQUE|

-31 FOLLOWING

2,924 FOLLOWERS

Simulation: A distributed counter



Demo: <https://softech-git.informatik.uni-kl.de/research/crdt-showcase>

Strong Eventual Consistency

- **All nodes that have received the same updates have the same state.**
- Changes made to a node propagate to all other nodes.
- Each node will **eventually** be consistent, but not necessarily immediately
- Conflicts arising from simultaneous changes at different nodes are automatically and deterministically resolved.

Conflict-free Replicated Data Types (CRDTs)

- Data consistency in distributed systems without a central conflict resolution mechanism
- Use data type semantics to resolve conflicts deterministically with predefined strategy
- Based on mathematical properties:
 - **Associativity**
 - **Commutativity**
 - **Idempotence**
- And some trick for symmetry breaking (arbitration order)

Classification of CRDT-Algorithms

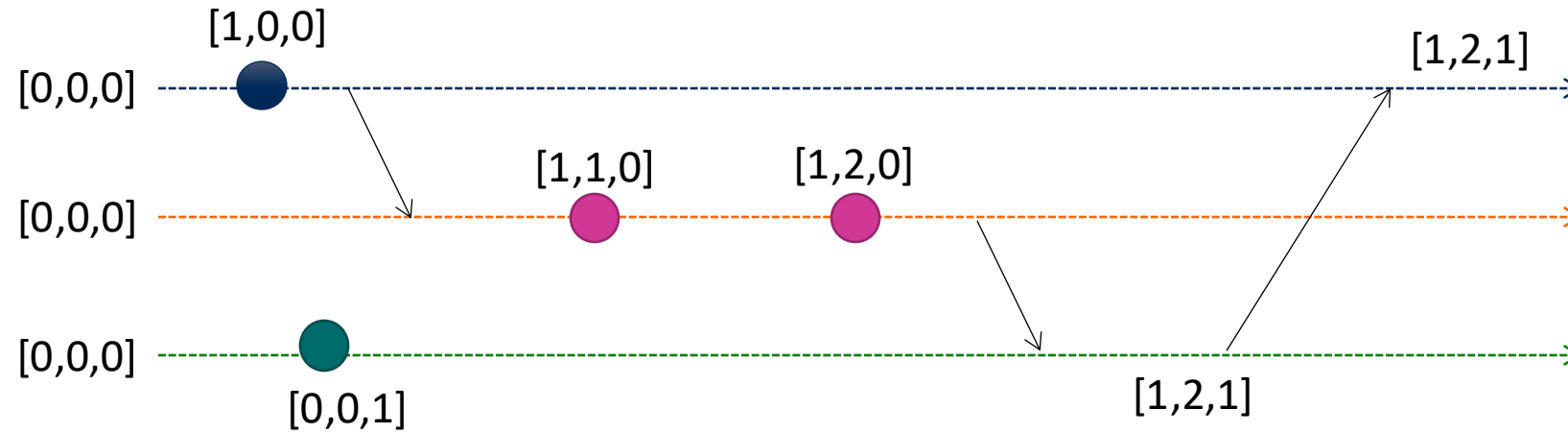
State-based

- Local states are exchanged and merged

Operation-Based

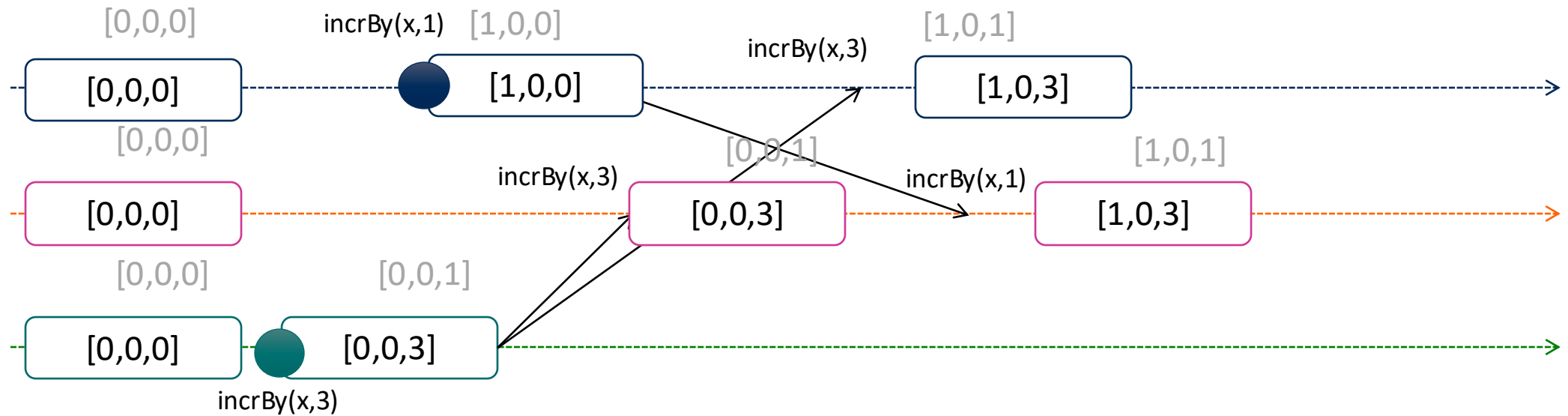
- Operations are forwarded to the other replicas and re-executed
- Requires deterministic operations and, in general, a partial order on the operations to maintain causal dependencies of operations.

Vector Clocks

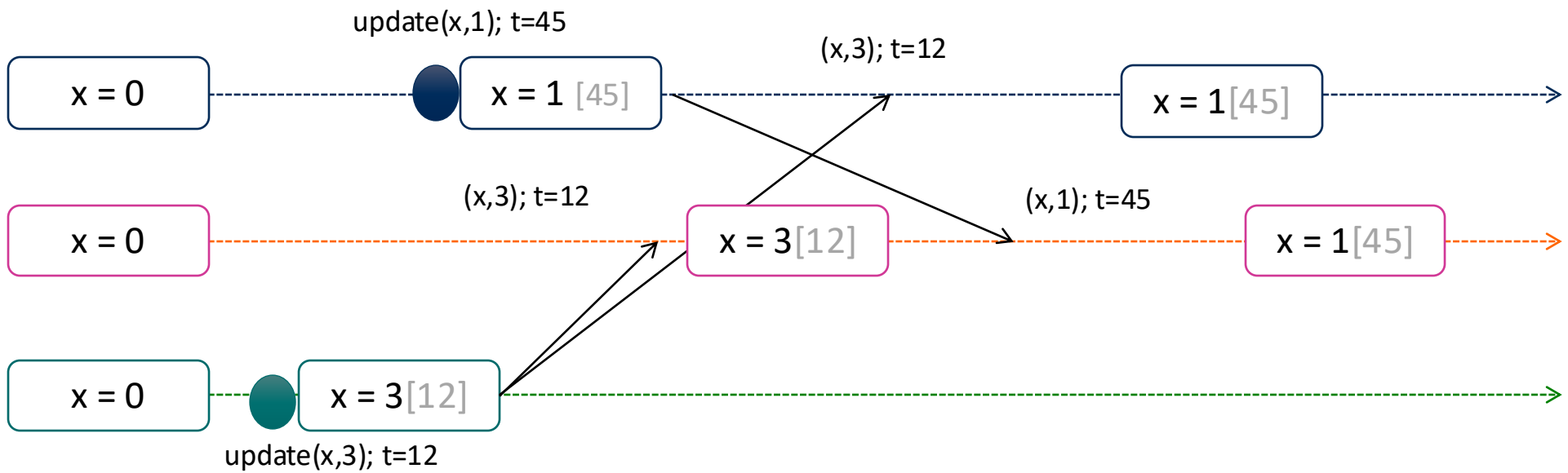


- $V_i[j] = \# \text{ events of node } j, \text{ observed by node } i$
- $V(a) < V(b) = \text{event } a \text{ happened before event } b$
- $V(a) < V(b) \wedge V(b) < V(a) = \text{events } a \text{ und } b \text{ are concurrent}$

State-based Counter



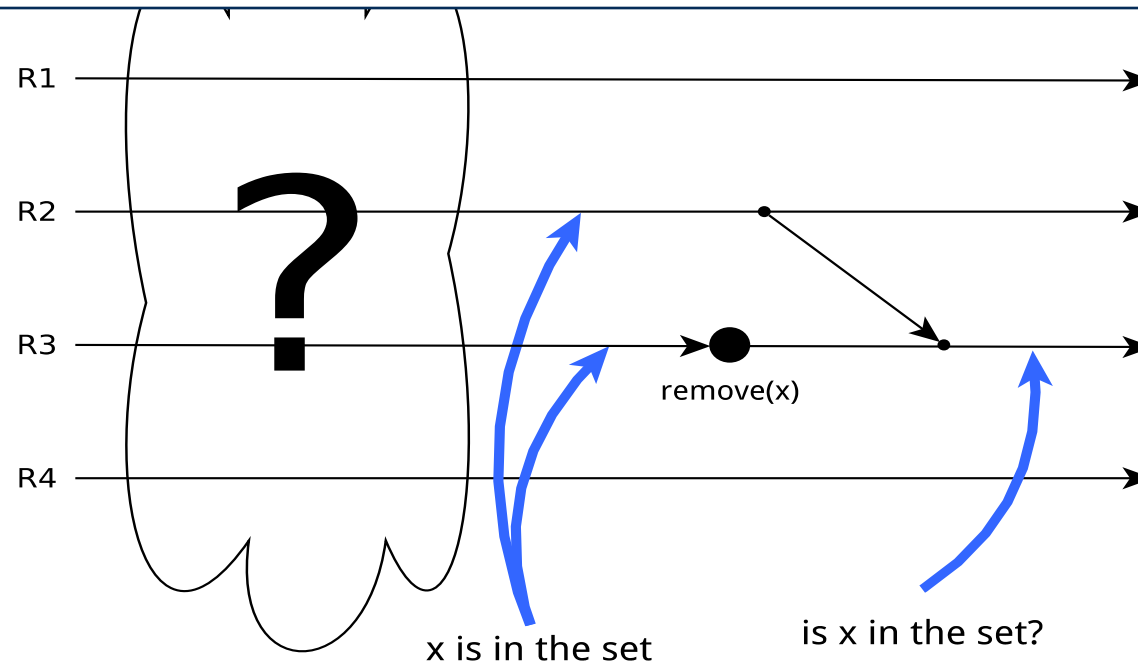
Last writer wins (LWW)



Add-wins (AW)

“Remove operation deletes only elements from the set that have been observed at the replica issuing the remove operation.

When concurrently adding the element (again), it will remain in the set.”



Replicated Sets with “interessanting” semantics

Grow-only set

- Only allows adding elements (no remove)
- Merge = Set union

2P-Set (Wuu & Bernstein PODC 1984)

- Two phases per element: Add an element, then remove the element
- Representation: Set of elements + set of tombstones
- Problem: Elements can never be added again

c-set (Sovran et al., SOSP 2011)

- Counts how often an element has been added and removed
- Element is contained if more adds than removes
- Problem: Might need to re-add element several times to “revive” it

A mathematician, a biologist, and a physicist are sitting at a street café, observing the house across the street.

First, they see two people enter the house.

Time passes. After a while, they notice three people coming out of the house.

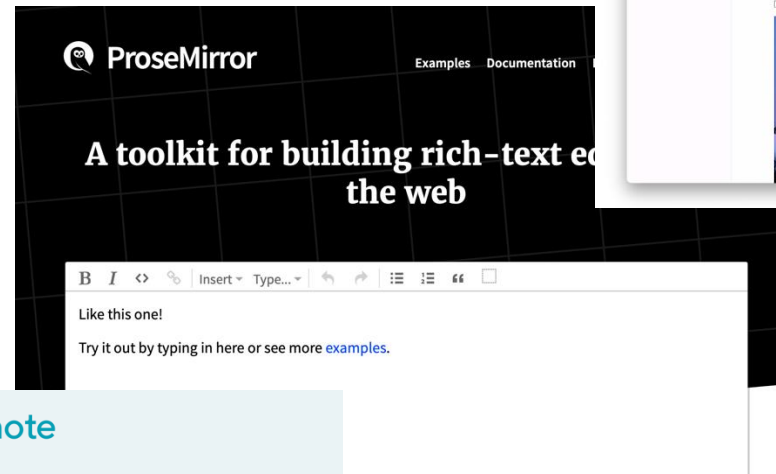
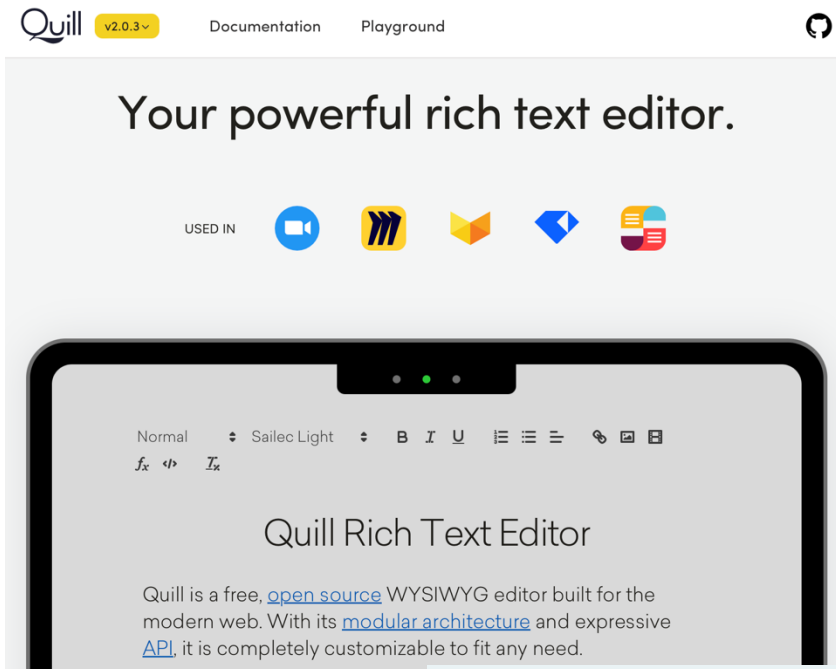
The physicist: "The measurement was not accurate."

The biologist: "They must have reproduced."

The mathematician: "If one more person enters, then the house will be empty again."

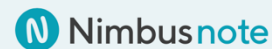
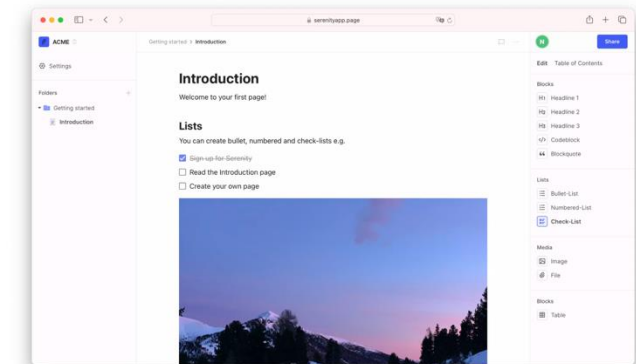
How to build Local-first Software

Collaborative text editors



Collaborate in real-time with ease

Intuitive rich-text editing with no conflict management needed while syncing in real-time.



One place to manage all your | projects

Transform information chaos from multiple sources into an organized workplace.
Empower yourself or your team to get things done faster with fewer iterations and meetings.

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Collaborative programming and development

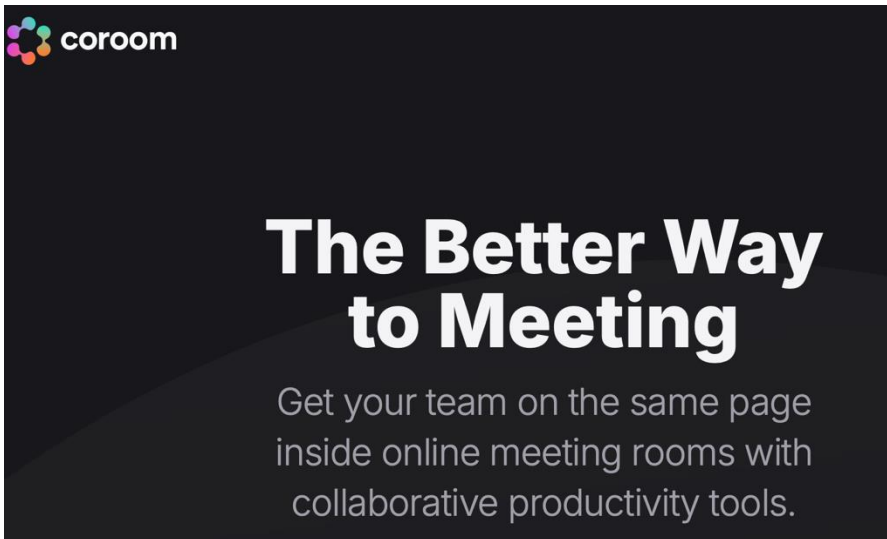
Welcome to JupyterLab Real-Time collaboration documentation!

From JupyterLab v4, file documents and notebooks have collaborative editing using the JupyterLab Real-Time Collaboration extension. Editors are not collaborative by default; to activate it, install the extension `jupyterlab-realtime-collaboration`.

Installation using mamba/conda:

Think bigger. Build faster.

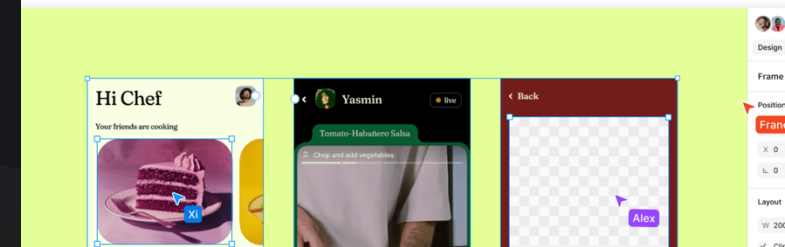
Figma helps design and development teams build great products, together.



coroom

The Better Way to Meeting

Get your team on the same page inside online meeting rooms with collaborative productivity tools.



Frameworks for CRDTs: Yjs

- Library of data types for collaborative applications
- Large ecosystem for persistence, update propagation etc.
- Supported datatypes:
Y.Map, Y.Text, Y.Array, Y.Xml
- Architecture agnostic (P2P, server-client)
- Featured in many rich-text editors
 - Offline mode
 - Version snapshots
 - Undo/Redo
 - Shared cursor positions

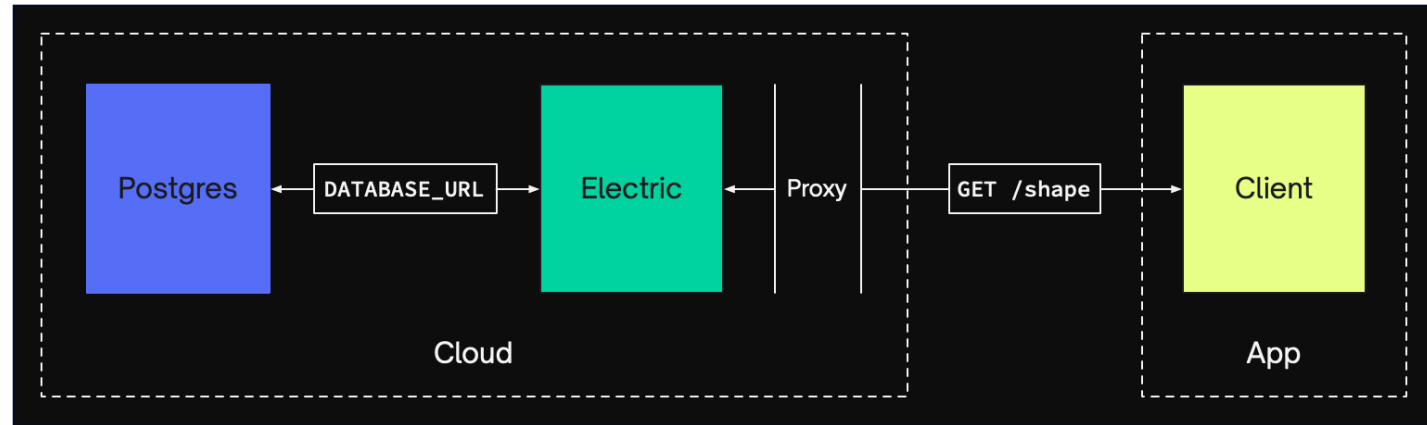


Frameworks for CRDTs: Automerge

- Library of data types for collaborative applications
- Solutions for persistence, update propagation, etc.
- Supported datatypes:
Documents (similar to JSON objects) with maps, lists, text and scalar types
- Architecture agnostic (P2P, server-client)
- Implementation in Rust with binding for JavaScript/WASM, Swift, ...



Sync services: Electric SQL

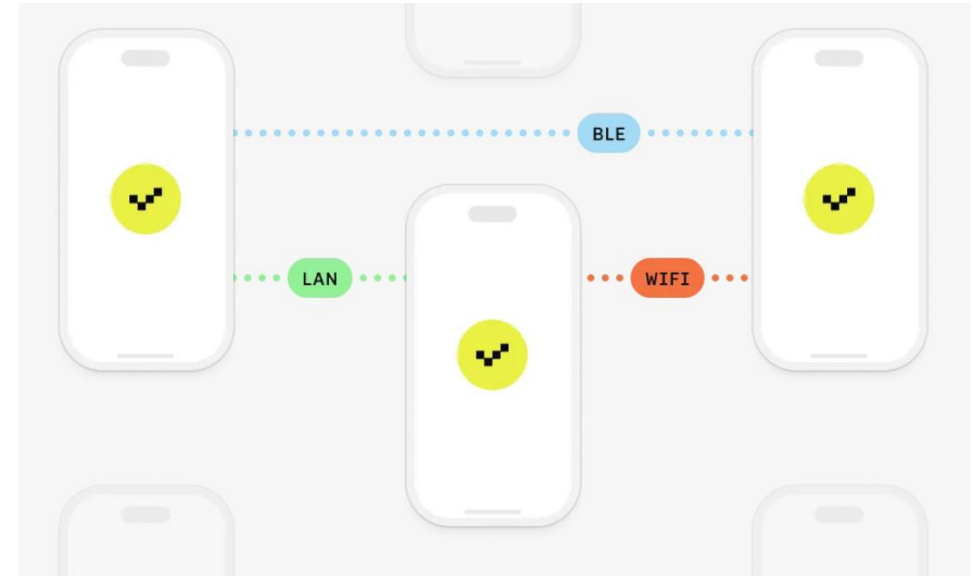


- Syncs subsets of Postgres data from cloud to local apps and services
- Built-in caching with distribution of data via CDNs
- State-transfer and cache invalidation with scalability and availability

Synchronization on the Edge: Ditto



- Focused on mobile and edge devices
- Direct P2P sync and opportunistic cloud sync
- Automatic device discovery and mesh networking
- Resilient



Teens' Lab Book

☰
📄
👤
💬
📅

Säure/Base-Titrationen Programm 1
 B1 Salzsäure mit Natronlauge gegen Indikator

?

ToolBox

- Text
- Tabelle
- Diagram
- Bild
- ToDo Liste
- HTML

Importieren/
Exportieren

Skript Labels

Notizen Labels

Tafel Labels

Ausloggen

Auswertung pH-Wert abhängige Enzymaktivität am Beispiel der alpha-Amylase

Aufgaben:

Sie haben in der simulierten Unterrichtsstunde "Quantifizierung von Stärkeabbau" mittels Handysphotometer die Stärkekonzentration nach dem enzymatischen Abbau durch alpha-Amylase bei einer Ausgangskonzentration von 0,1 g/l für verschiedene pH-Werte gemessen. Eine beispielhafte Messung mit entsprechenden Messwerten haben Sie in der Mail erhalten.

- Erstellen Sie auf Ihrem persönlichen Board (📌) zunächst eine Tabelle und tragen in die erste Spalte die pH-Werte und in eine zweite Spalte die gemessenen Konzentrationen in [g/l] ein. Geben Sie den Spalten zusätzlich passende Überschriften und ändern Sie das Label der Komponente so ab, dass dieses eine nachvollziehbare Beschriftung bekommt. Beachten Sie, dass das Tool ein Dezimalkomma mit einem Punkt darstellt. Daher muss beispielsweise 0,1 als 0.1 eingegeben werden!
- Aus den ermittelten Messwerten berechnen Sie in einer neuen Spalte die Konzentrationsdifferenz in [g/l] als Maß für die Enzymaktivität. Hierzu müssen jeweils die Differenzen aus Kontrollwert und gemessener Konzentration gebildet werden. Verwenden Sie hierzu die Differenzfunktion für verschiedene Zellen (z.B. "(A8-A1)" Klammern sind hier wichtig!). Passen Sie ebenfalls die Spaltenüberschrift entsprechend an.
- Legen Sie in der Tabelle eine weitere Spalte an und tragen Sie die berechneten Werte hier ein. Dies ist notwendig, da die App noch keine berechneten Werte darstellen kann. Stellen Sie anschließend die Konzentrationsdifferenz in Abhängigkeit vom pH-Wert in geeigneter Form in einem Diagramm dar. Ändern Sie auch für diese Komponente das Label sinnvoll ab. Achten Sie auf korrekte Beschriftung des Diagramms und begründen Sie kurz die Wahl des Diagrammtyps in einem separaten Textfeld.
- Wechseln Sie zur Tafelansicht (📊) und vergleichen Sie Ihr Diagramm hinsichtlich fachwissenschaftlich korrekter Darstellung mit dem Diagramm auf der Tafel. Fassen Sie Ihre Ergebnisse stichpunktartig in einer separaten Textbox auf Ihrem persönlichen Board (📌) zusammen.
- Laden Sie das beigefügte Foto auf Ihr persönliches Board (📌) hoch und ändern das Label der Komponente sinnvoll ab.
- Füllen Sie den Fragebogen unter <https://didaktik.biologie.uni-kl.de/limesurvey/index.php/921518?lang=de> aus.

Skript

0.40x

+

-

↺

👁

📄

Tabelle: 19/07/23, 11:00-11:07

	XYZ	XYZ	XYZ	XYZ
1	4.7	0.0632		0.0525
2	5.4	0.0319		0.0281
3	5.8	0.0158		0.0194
4				
5	6.5	0.0101		0.0237
6	7.0	0.013		0.0285
7	7.7	0.042		0.0493
8	8.5	0.0714		0.0883
9	KT	0.1048		0.115

Diagram: 19/07/23, 12:41-11:07

Diagram: 23/10/23, 16:56-11:11

Tafel

0.74x

+

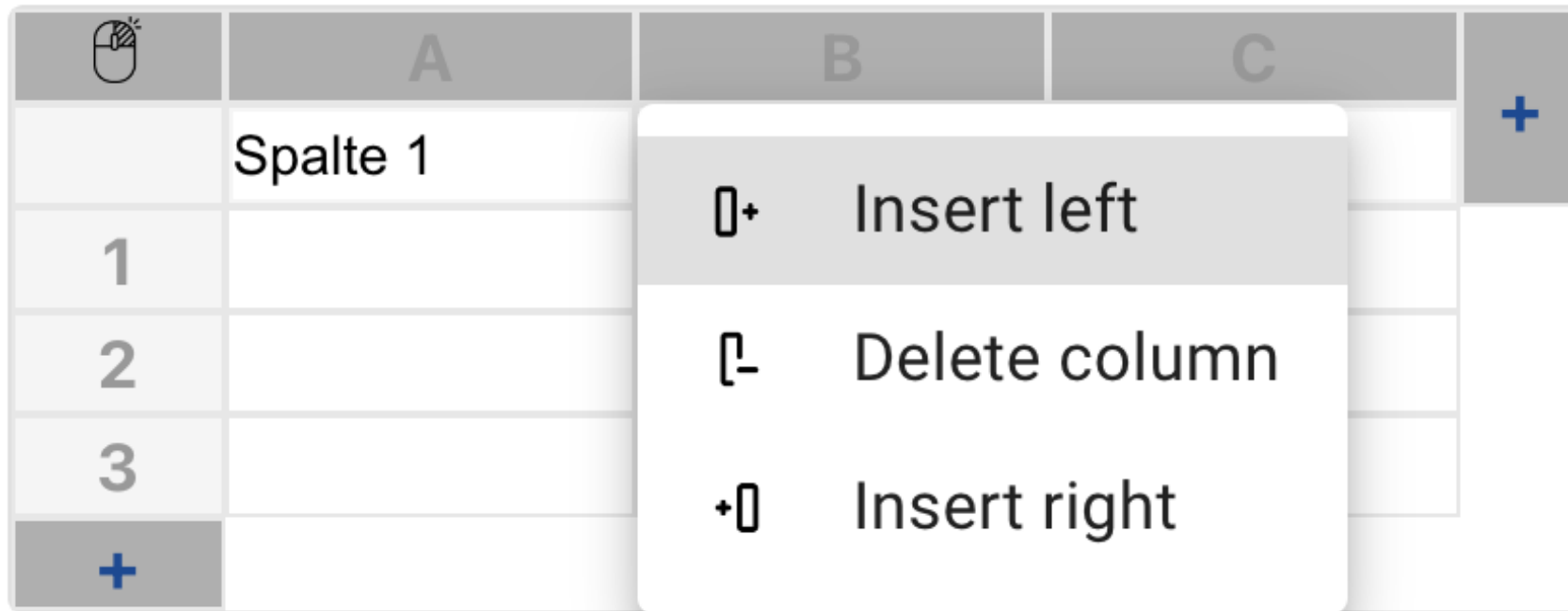
-

↺

👁

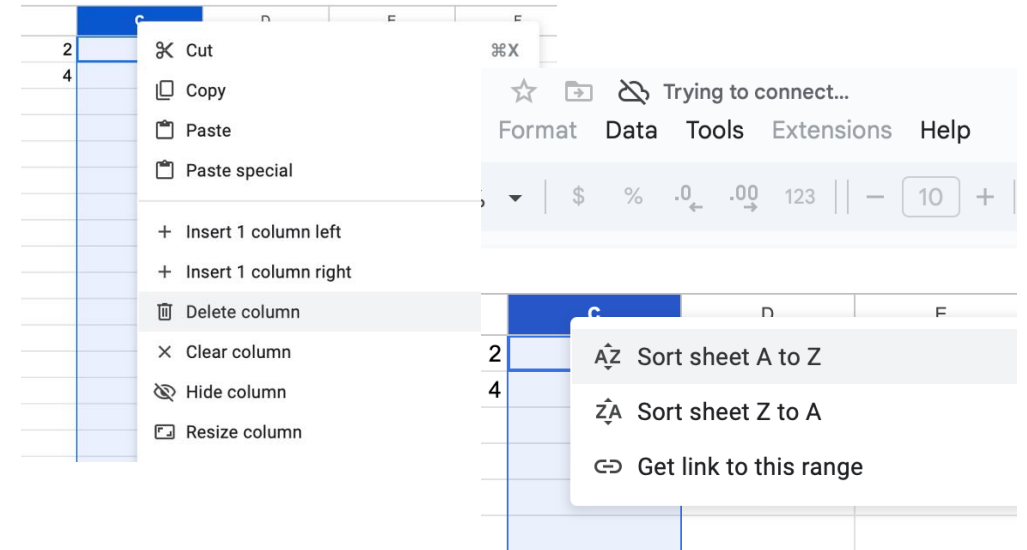
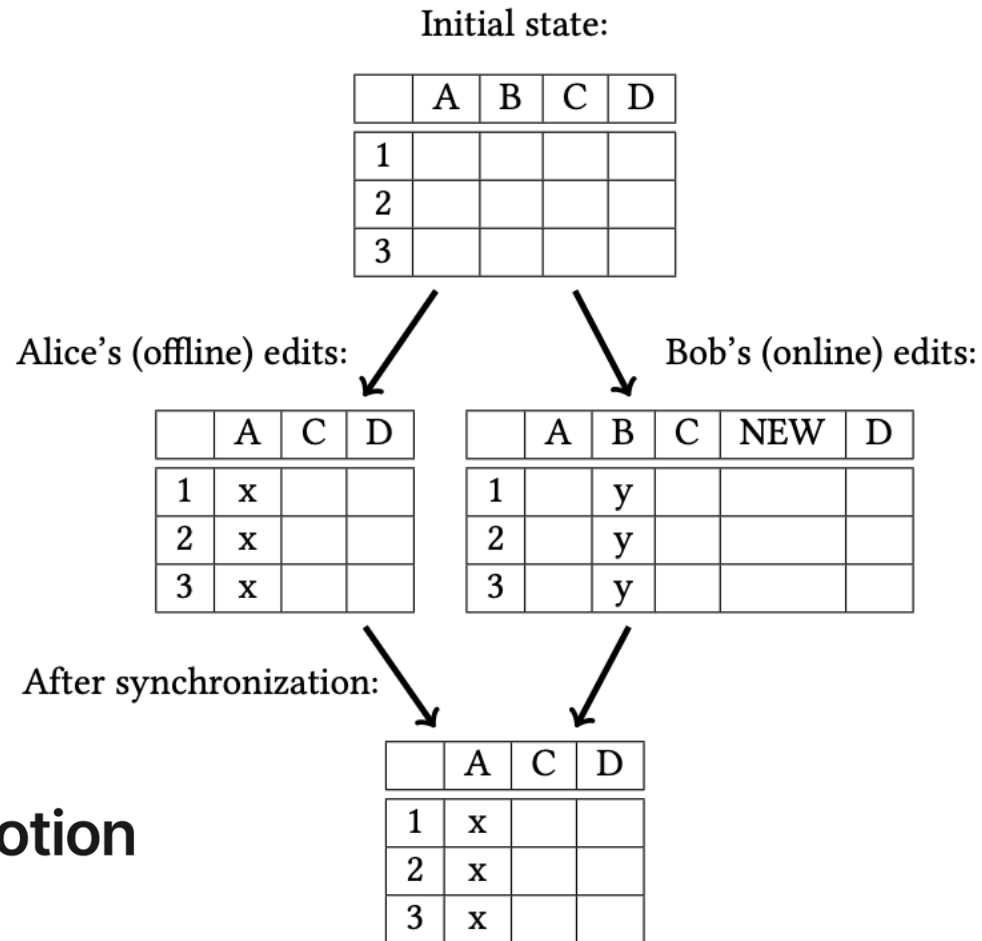
📄

Collaborative Spreadsheets



Conflict resolution in collaborative spreadsheets

[Yanakieva et al. 2022]



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Which operations conflict?

	Insert column	Delete column	Insert row	Delete row	Edit cell
Edit cell		If same column		If same row	If same cell
Delete row				If same row	
Insert row			If same position		
Delete column		If same cell			
Insert column	If same position				

	Non-commutative
	Commutative
	Implementation dependent

Design decisions: Editing the same cell

Which value should the cell contain if two users edit the cell content concurrently?

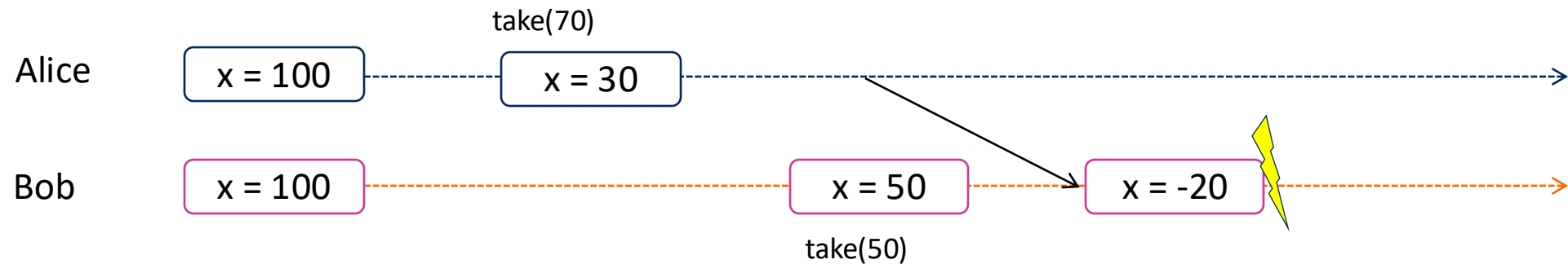
- The largest value
- The smallest value
- Both values appended
- None of the values
- A choice of both values
- The value last written / arbitrarily one of them
- Alice write always overwrites Bob's value
- ...

**What are the limits of Local-First
(currently)?**

Data invariants that require strong consistency

Examples

- Transfer between bank accounts
- No overselling of tickets
- Keeping graphs cycle-free
 - unless you find a symmetry-breaking insight



Consensus, leases, ...

Research on Invariants

- Invariants in Graph CRDTs [Borth et al., PaPoC'25]
- Maintaining integrity constraints in replicated relational databases with integrity constraints [Ignat et al., DAIS'24]
- Lore: Programming models for verifiable safe local-first software [Haas et al., TOPLAS'24]
- Hamsaz: Replication Coordination Analysis and Synthesis [Houshmand and Lesani, POPL'19]
- Sequential Programming for Replicated Data Stores [Lewchenko et al., ICFP'19]
- IPA: Invariant-Preserving Applications for Weakly Consistent Replicated Databases [Balegas et al., VLDB'18]
- And many more ...

Challenges and Open Problems

Access control:

When are dynamically changing access control policies checked and enforced?

Local-first transformation:

Can existing applications be automagically made collaborative and local-first?

Standardized data formats:

Which data formats are long-lived and independent? What meta-data is required?

Causal dependencies:

How can operations be undone/redone without violating dependencies?

Summary: Local-First Software

- „If it's local-only, it's not local-first.“
- “If it doesn't work with the wifi off, it's not local-first.“
- „If it doesn't work if the app developer goes out of business and shuts down their servers, it's not local-first.“

Martin Kleppmann
(Local-first Conference, Berlin 2024)

Happy and Successful Sync'ing!



**LOCAL
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BERLIN, GERMANY
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LOCAL-FIRST CONF 2025

Thank you!

